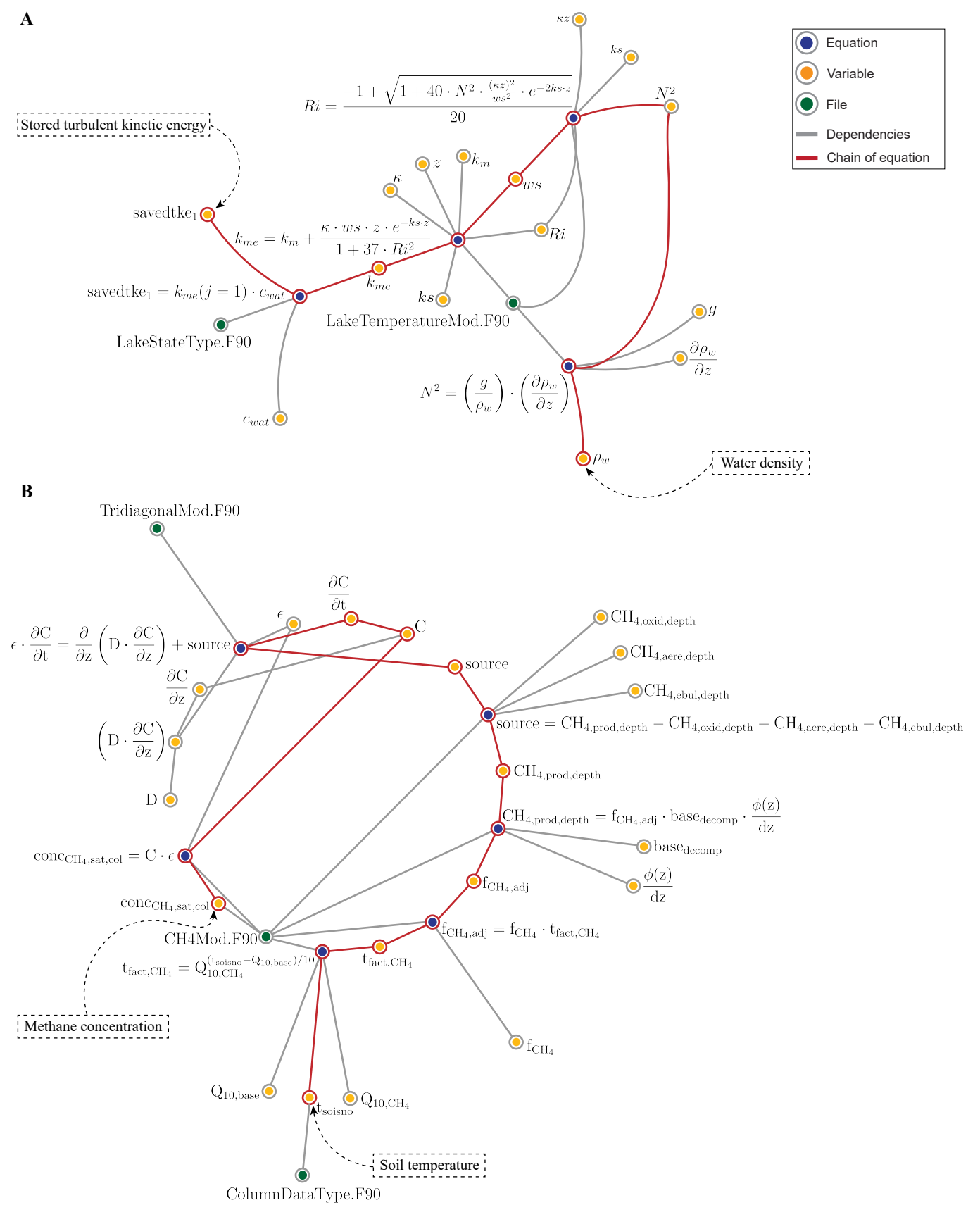

A
Equation
Variable
File
Dependencies
Chain of equation
Stored turbulent kinetic energy
savedtke$_1$
$k_{me} = k_m + \frac{\kappa \cdot ws \cdot z \cdot e^{-ks \cdot z}}{1 + 37 \cdot Ri^2}$
$Ri = \frac{-1 + \sqrt{1 + 40 \cdot N^2 \cdot \frac{(\kappa z)^2}{ws^2} \cdot e^{-2ks \cdot z}}}{20}$
κz
ks
N^2
z
k_m
κ
ws
Ri
k_{me}
ks
savedtke$_1 = k_{me}(j = 1) \cdot c_{wat}$
LakeStateType.F90
LakeTemperatureMod.F90
c_{wat}
g
$\frac{\partial \rho_w}{\partial z}$
$N^2 = \left(\frac{g}{\rho_w}\right) \cdot \left(\frac{\partial \rho_w}{\partial z}\right)$
ρ_w
Water density
B
TridiagonalMod.F90
$\epsilon \cdot \frac{\partial C}{\partial t} = \frac{\partial}{\partial z}\left(D \cdot \frac{\partial C}{\partial z}\right) + source$
$\frac{\partial C}{\partial t}$
ϵ
C
$\frac{\partial C}{\partial z}$
$\left(D \cdot \frac{\partial C}{\partial z}\right)$
D
source
$CH_{4,oxid,depth}$
$CH_{4,aere,depth}$
$CH_{4,ebul,depth}$
$source = CH_{4,prod,depth} - CH_{4,oxid,depth} - CH_{4,aere,depth} - CH_{4,ebul,depth}$
$CH_{4,prod,depth}$
$CH_{4,prod,depth} = f_{CH_4,adj} \cdot base_{decomp} \cdot \frac{\phi(z)}{dz}$
$base_{decomp}$
$\frac{\phi(z)}{dz}$
$conc_{CH_4,sat,col} = C \cdot \epsilon$
$conc_{CH_4,sat,col}$
CH4Mod.F90
$f_{CH_4,adj}$
$f_{CH_4,adj} = f_{CH_4} \cdot t_{fact,CH_4}$
$t_{fact,CH_4} = Q_{10,CH_4}^{(t_{soisno} - Q_{10,base})/10}$
t_{fact,CH_4}
f_{CH_4}
Methane concentration
$Q_{10,base}$
t_{soisno}
Q_{10,CH_4}
Soil temperature
ColumnDataType.F90